Project Design Phase
Proposed Solution Template

Date	30 September 2026
Team ID	SWTID-2026-2365
Project Name	AI FitTrack API – Fitness Tracking API with Google Gemini AI
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Fitness enthusiasts, especially beginners, have no single, secure place to log their workouts and no affordable way to get personalised guidance. Records are scattered across notebooks and apps, and generic plans ignore a person's age, goal and experience, so many users lose motivation and quit.
2.	Idea / Solution description	AI FitTrack API is a RESTful back end built with Node.js, Express.js and MongoDB. Users register and log in with JWT authentication, record workouts (name, category, duration, calories, date), search them by name, category or date, and ask Google Gemini AI for a personalised workout recommendation and a short fitness insight based on their statistics.
3.	Novelty / Uniqueness	One secured API combines a workout tracker with generative AI. Prompts are designed to return short plain-text answers that any client can show directly. Search accepts name, category or date in a single query, and every workout is strictly scoped to its owner. The same API can power web, mobile or chatbot clients.
4.	Social Impact / Customer Satisfaction	Gives beginners affordable, personalised guidance that would otherwise need a paid trainer and encourages regular exercise and a healthier lifestyle. A single searchable history and simple AI feedback keep users informed and motivated, and passwords and data are protected by hashing and token-based access.

5.	Business Model (Revenue Model)	Freemium model: core tracking free, with a premium plan for unlimited AI recommendations and advanced insights. Additional revenue from API subscriptions for gyms, wellness apps and corporate wellness programmes (pay-per-use or monthly quota), and partnerships with trainers and nutrition brands.
6.	Scalability of the Solution	The API is stateless (JWT), so several instances can run behind a load balancer. MongoDB Atlas supports replica sets and sharding, and the modular MVC structure allows new modules to be added. The AI calls sit in a separate service layer, so caching, queuing or a change of model can be added without touching other code.